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Curriculum-Based Measurement of Oral Reading: Standard Errors Associated With Progress Monitoring Outcomes From DIBELS, AIMSweb, and an Experimental Passage Set

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Abstract. There are relatively few studies that evaluate the quality of progress monitoring estimates derived from curriculum-based measurement of reading. Those studies that are published provide initial evidence for relatively large magnitudes of standard error relative to the expected magnitude of weekly growth. A major contributor to the observed magnitudes of standard error is the inconsistency of passage difficulty within progress monitoring passage sets. The purpose of the current study was to evaluate and estimate the magnitudes of standard error across an experimental passage set referred to as the Formative Assessment Instrumentation and Procedures for Reading (FAIP-R) and two commercially available passage sets (AIMSweb and Dynamic Indicators of Basic Early Literacy Skills [DIBELS]). Each passage set was administered twice weekly to 68 students. Results indicated significant differences in intercept, weekly growth, and standard error. Estimates of standard error were smallest in magnitude for the FAIP-R passage set followed by the AIMSweb and then DIBELS passage sets. Implications for choosing a progress monitoring passage set and estimating individual student growth are discussed.

In the late 1970s and early 1980s, Stan Deno and colleagues developed the procedures for curriculum-based measurement of oral reading (CBM-R) to enable teachers to systematically monitor and evaluate the effects of instruction on student performance (Deno, 1985; Deno, Marston, & Tindal, 1985; Deno, Mirkin, & Chiang, 1982). Research suggests that students are likely to make greater academic gains if their teachers use CBM-R in conducting systematic formative evaluations to determine when and if instructional modifications are needed (Fuchs, Deno, & Mirkin, 1984; Fuchs & Fuchs, 1986; Fuchs, Fuchs, Hamlett, & Allinder, 1991). Systematic formative evaluation using CBM-R involves administering passages twice weekly, weekly, or monthly, and plotting a student's observed

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words read correctly per minute (WRCM) in time series fashion. CBM-R allows for the depiction of student growth through the plotting of data in time series fashion because it departs from traditional psychometrics as it "integrates the concepts of standardized measurement and traditional reliability and validity with features from behavioral and observational assessment methodology" (Deno, Fuchs, Marston, & Shin, 2001, p. 508).

Unfortunately, with few exceptions (e.g., Christ & Ardoin, 2009; Poncy, Skinner, & Axtell, 2005), the majority of research evaluating CBM-R has failed to examine its technical features beyond its reliability and validity, which is necessary but not sufficient for measures used to depict student growth (Deno et al., 2001; Francis et al., 2008). Francis et al. (2008) suggested that measures used to depict growth must have a sufficient number of alternate forms and the construct and difficulty level of the forms must remain constant across measurement occasions. When there is variability in the construct measured or the difficulty of the probes, variability appears in student performance that is not a function of changes in skills, resulting in less than accurate depictions of student growth.

Initially the multiple forms that made up CBM-R passage sets for progress monitoring were developed by randomly selecting passages from students' curricula. This method, however, was discovered to be flawed because of considerable variability in the difficulty of text within curricula (Fuchs & Deno, 1992, 1994). In an effort to provide schools with sets of standardized controlled passages, researchers have developed passage sets consisting of probes with difficulty levels equated using readability formulas. Readability formulas assess the difficulty of passages based on factors such as the number of single-syllable words per 100 words, syllables per 100 words, or percentage of words in the passages not found in a word list (Ardoin, Suldo, Witt, Aldrich, & McDonald, 2005). According to the Dynamic Indicators of Basic Early Literacy Skills (DIBELS) technical manual, each step in the development of the passage sets involved using readability formulas as the sole means of equating passages (Good & Kaminski, 2002). Procedures employed in the development of AIMSweb passage sets were similar; however, in an effort to further evaluate passage equivalence, approximately 20 students read each passage to eliminate passages seemingly too easy or difficult (Howe & Shinn, 2002).

Despite a heavy reliance on readability formulas, multiple studies indicate that readability formulas are not adequate for equating passages (Poncy et al., 2005). Studies examining the sensitivity of CBM-R progress monitoring procedures have repeatedly shown that the formulas fail to yield passages of equivalent difficulty (Dunn & Eckert, 2002; Hintze, Shapiro, & Lutz, 1994). Dunn and Eckert (2002) reported that variability in individual student performance as a function of uncontrolled passage difficulty was so great that time failed to account for variability in students' WRCM. Hintze et al. (1994) reported that estimates of readability provided by the readability formulas used in their study were uniformly low and nonsignificant "across all possible instructional and measurement arrangements" (p. 194). Furthermore, research directly evaluating the utility of readability formulas for selecting equivalent passages revealed minimal to no relationship between students' oral reading fluency and the readability formulas typically used within the CBM-R literature (Ardoin et al., 2005; Christ & Ardoin, 2009; Compton, Appleton, & Hosp, 2004; Poncy et al., 2005).

Variability in passage difficulty and the resulting lack of accuracy of estimates in student growth (Christ & Coolong-Chaffin, 2007) might be surprising to those who are familiar with the many optimistic statements within the professional literature regarding CBM-R. Unfortunately, few studies have addressed the issue of measurement consistency and error within the context of CBM-R progress monitoring at the individual child level. The few published studies that report rates of expected growth provide standard errors for data at the group level (Deno et al., 2001; Fuchs, Fuchs, Hamlett, Walz, & Germann, 1993) as opposed to the individual student level. Evidence of technical adequacy based on group level data

has limited applicability to individual assessment data (Fuchs, Tindal, & Deno, 1984). Given that, in practice, progress monitoring data and CBM-R slopes are evaluated and interpreted at the level of the individual student, it is essential that we attend to the reliability and accuracy of estimates of individual student growth as opposed to only at the group level.

Christ (2006) conducted one of the few studies examining the stability of progress monitoring outcomes and CBM-R slopes for individual students. Published estimates for standard error of the estimate (*SEE*) were used to estimate the standard error of slopes (*SEb*). Even when using the smallest reported values of *SEE* and 10 weeks of twice-weekly progress monitoring data, which is more data points than often is collected before making instructional decisions in practice, estimates of *SEb* were sizable (0.78) when compared to the expected rate of weekly growth (1–2 WRCM per week; Deno et al., 2001). Such results have led some to advocate for research to improve the quality of CBM-R instrumentation as variability in passage difficulty "detracts from accurately representing growth, it undermines timely instructional decision making and timely evaluation of RTI" (Jenkins, Zumeta, Dupree, & Johnson, 2005, pp. 245–246).

Because of concerns over the variability in passage difficulty within published passage sets and the resultant error in progress monitoring data, Christ and Ardoin (2009) explored alternative procedures for selecting equivalent passages. After developing 50 third-grade CBM-R passages, 46 students read each of the developed passages within a 1-week period. Sampling procedures were then used to construct four sets consisting of 20 passages each: (a) random, (b) readability, (c) mean, and (d) Euclidean distance. The procedures used to construct the random passage set were consistent with published recommendations (Shapiro, 1996). Procedures used to construct the readability passage set were consistent with previous applications (Hintze & Christ, 2004) in which a readability formula was used to screen and select passages that fell within a predetermined readability range. The procedures used to construct the mean passage set were consistent with recent published recommendations for screening and selecting passages with reference to performance means (Poncy et al., 2005). Finally, a Euclidean distance passage set was selected as a novel alternative to all other previously published recommendations. Euclidean distance is a type of distance measure that can be used to group (similar) observations/passages and identify outliers (dissimilar) observations/passages. This procedure, and other similar procedures (e.g., Mahalanobis distance), are frequently employed in classification (e.g., cluster analysis) and to detect outlier values for modeling (e.g., linear regression) and multivariate analysis. Specifically, determining the Euclidean distance between data points involves calculating the square root of the sum of the squared differences between corresponding values.

In an effort to examine the proportion of variance associated with the target of measurement (student/person), passages, and residual error, Christ and Ardoin (2009) employed generalizability studies (Brennan, 2001). Results indicated that the proportion of variance associated with alternate passages was only 1% for the Euclidean distance passage set and as high as 10% for the random passage set. These data suggested that the Euclidean distance procedure for selecting passages might be useful for reducing measurement error observed when monitoring student progress, thus increasing the reliability and accuracy of estimates of individual student growth attained through CBM-R progress monitoring.

Purpose

Existing research suggests there is substantial error in CBM-R progress monitoring, primarily due to variability in passage difficulty, resulting in inaccurate estimates of individual student growth. The purpose of the current study was therefore to evaluate measurement error across a field tested passage set referred to as the Formative Assessment Instrumentation and Procedures for Reading (FAIP-R) and two commercially available

passage sets (AIMSweb and DIBELS). Equivalence of FAIP-R was determined using students' reading of a larger set of passages and then selecting those passages with the smallest Euclidean distance. The equivalence of passages within AIMSweb and DIBELS passage sets were based primarily upon readability formulas. Given evidence that the Euclidean distance procedure reduced variance associated with alternate passages (Christ & Ardoin, 2009), it was expected that passages identified as having equivalent difficulty by the Euclidean distance procedure (FAIP-R passages) would reduce observed measurement error as compared to passage sets based on similar readability estimates (i.e., AIMSweb and DIBELS). In addition, it was hypothesized that the AIMSweb passage set would result in significantly less measurement error than the DIBELS passage set because AIMSweb passages were equated using both readability formulas and observed performance of a small sample of students at each grade level. Two forms of measurement error were examined: *SEb* and *SEE*. Although *SEE* is the form of measurement error typically reported in the evaluation of an instrument, within the context of evaluating an individual student's progress, we believe *SEb* to be more informative as it provides information regarding the accuracy of obtained slopes. Previous studies have found that CBM-R progress monitoring results in *SEb* that are larger than expected rates of gain, resulting in noninterpretable estimates of individual student's rate of growth (Christ, 2006; Francis et al., 2008). An additional goal of this study was to examine differences in estimates of intercept and student growth as a function of passage set.

Method

Participants and Setting

Participants included 28 second-grade and 40 third-grade students with a mean age of 8 years 2 months (range = 7 years 0 months to 9 years 6 months). Second- and third-grade students were chosen for this study because students at these grade levels read the original set of FAIP-R passages from which the equated passages were selected (Christ & Ardoin, 2009). The majority of students were Caucasian (73%), with the remaining students being identified as either African American (21%) or other (6%). Students were enrolled in one of two elementary schools located in the southeast. Forty-eight of the students were enrolled in School A, which served students in kindergarten through fifth grade. Within School A, 74% of the students were Caucasian, 19% African American, 6% Other; 19% of the student body received free or reduced-price lunch. The remaining 20 students attended School B. This school also served students in kindergarten through fifth grade, of which 74% were Caucasian, 15% African American, 11% other, and 32% received free or reduced-price lunch. For 32 students, sessions were conducted at a time specified by their teachers, with progress monitoring sessions for the remaining 36 students being conducted during an after school program offered by the school. The after school program was not specific to students with any identified deficits or instructional needs.

Participants were recruited through one of two methods. All second- and third-grade students who attended the after school program were sent consent forms informing the parents of the study and its benefits and risks. Approximately 70% of the parents agreed to allow their child to participate. For the participating children, teachers sent consent forms to the parents of 5 students in their classroom who had slightly below-average to average reading skills, had limited behavioral issues, and regularly attended school. Teachers were not provided with any additional criteria regarding how to choose students with below-average reading skills. Although they did have access to CBM-R universal screening data, we cannot be certain exactly how teachers selected these students. Approximately 80% of these consent forms were returned.

Sessions lasted approximately 10 min. For those students who were enrolled in the after school program, sessions took place in a classroom where other students were either working or being assessed using similar procedures. When other assessment sessions were

ongoing, students were faced away from each other with a minimum of 10 feet between students. The sessions of those students who were not enrolled in the after school program occurred in a teacher workroom or a hallway adjacent to students' classrooms.

Materials

Three passage sets consisting of 20 passages each were used in monitoring the progress of all participants. The FAIP-R passage set developed by the authors was comprised of 20 passages selected based upon data reported within Christ and Ardoin (2009). In this study 50 passages were administered to 46 second- and third-grade students within a week's period of time. For the purpose of the current study, passages were rank ordered by grade according to their Euclidean distance as calculated using SPSS. The 26 passages across second and third grade with the smallest Euclidean distance yielded 20 passages common across both grades. Those passages were arranged for progress monitoring administration beginning with the median estimated Euclidean distance then alternating between the lesser and greater estimates. That is, the 11th rank passage was administered first followed by the 10th, 12th, 9th, 13th, 8th until the final two passages were the 1st and 20th ranked passages, respectively.

Students' progress was also monitored using AIMSweb and DIBELS progress monitoring passages. The first 20 passages identified as the AIMSweb third-grade progress monitoring passages and the 20 DIBELS third-grade progress monitoring passages were administered to students in the order specified by each company.

Procedures

An effort was made to conduct assessments twice per academic week (Monday to Friday) with 1 day between sessions (e.g., Monday and Wednesday). On occasions when unscheduled absences occurred, sessions were made up when the student returned to school, resulting in some assessments being conducted with either 0 days or 2 days between sessions. Similarly, when school vacations or field trips were scheduled, sessions were conducted with either 0 or 2 days between sessions. On average, students read all passages within 12 weeks (range = 11–15 weeks).

During each assessment, students read one passage from each of the three passage sets (FAIP-R, AIMSweb, and DIBELS). The order of administration was counterbalanced within and across students. Thus, although every student read FAIP-R progress monitoring passage 5, AIMSweb progress monitoring passage 5, and DIBELS progress monitoring passage 5 during the fifth progress monitoring session, the order in which they read these three passages differed. Order of administration was determined for each student before commencement of the study.

Standard CBM-R instructions were provided to students before the first three passages read on Day 1 and before the first passage administered during all subsequent sessions. During Sessions 2–20, before reading the second and third passages, students were told "same instructions as before." In addition to informing students of the instructions each day, students were also informed before reading each passage that they could earn $0.05 for exceeding a predetermined score. On the first day students were told that they could earn $0.05 for doing their best reading. All students were provided with a nickel for each of these readings. During the second session, students were informed that if they exceeded their median score from the first day they could earn $0.05. After Sessions 1 and 2, the mean score from these two sessions was determined and for the next two sessions the student had to have 2 WRCM greater than the mean from the first two sessions. Thereafter, 2 WRCM were added after every two sessions, resulting in the criterion of the last two sessions being 10 words greater than the mean WRCM across the initial two sessions. After reading each passage, students were told the WRCM they achieved, whether they earned the $0.05, and provided with the $0.05 if earned. These procedures were employed to increase the probability of students giving their best effort across the 20 progress monitoring sessions.

At the conclusion of the study, ordinary least square regression was used to calculate an intercept and slope for each passage set. Slopes were each multiplied by 7 to transfer calculated slopes into an estimate of weekly gain in words read correctly (Hintze & Christ, 2004). Likewise, each calculated *SEb* was multiplied by 7 (Christ & Coolong-Chaffin, 2007).

Procedural Integrity and Interrater Reliability

Experimenters were trained in administration and scoring procedures during two 45-min sessions. Before conducting assessments, examiners had to achieve 100% procedural integrity and reliability with the first author during three consecutive assessments. During each universal screening, examiners used procedural checklists to ensure that they provided students with correct instructions and as a reminder of scoring rules.

Procedural integrity and interrater agreement were assessed on 16% of the sessions. Independent raters listened to tape-recorded sessions while scoring a reading probe and recording whether the examiner adhered to protocol. Procedural integrity was calculated by dividing the number of completed steps by the total number of steps and multiplying by 100%. The protocol for each day was divided into 17 steps (e.g., reading standardized instructions, administering passage according to prespecified order). Mean procedural integrity was 99% (range: 71%–100%). Procedural integrity fell below 100% on only 25 of 1360 sessions assessed.

Interrater agreement was assessed by comparing each word the examiner recorded as correct and incorrect to each word the independent rater scored as correct and incorrect. The number of words that both the examiner and rater agreed was correct and incorrect was divided by the total number of words on which they agreed and disagreed and multiplied by 100%. Interrater agreement was assessed on 16% of passages read and resulted in a mean interrater agreement of 99% (range = 87%–100%).

Results

Table 1 presents descriptive statistics for each dependent variable within each of the three passage sets. The characteristics of the data set were consistent with the assumptions of both multivariate and univariate analysis.

To assess the effects of passage sets on estimates of linear growth, a multivariate analysis of variance repeated across three passage sets and four measures of linear growth (intercept, weekly growth, weekly *SEb*, and *SEE*) was conducted. Consistent with recommendations in the literature (Stevens, 2002; Tabachnick & Fidell, 2001), analyses progressed from multivariate omnibus to univariate post hoc tests. Multivariate analysis was used to test for statistically significant differences across passage sets for the multivariate combination of dependent measures using Wilks's criterion, $F(8, 262) = 9060$, $p < .05$, $\eta^2 = .73$. After statistically significant differences and large effect sizes were observed, follow-up univariate analyses were conducted to test for statistically significant effects (using Greenhouse-Geisser) across passage sets for each of the four dependent measures. There were statistically significant differences between passage sets for measures of intercept, $F(1.9, 131.3) = 167.5$, $p < .05$, $\eta^2 = .71$; weekly growth, $F(2.0, 133.8) = 62.7$, $p < .05$, $\eta^2 = .48$; weekly SEb, $F(1.8, 125.5) = 67.08$, $p < .05$, $\eta^2 = .50$; and SEE, $F(1.9, 128.1) = 71.4$, $p < .05$, $\eta^2 = .52$. The magnitude of each effect size was within the medium (>.30) to large range (>.50; Cohen, 1988). Marginal means and corresponding standard errors are presented in Table 2 along with the results of paired contrasts to test for statistically significant differences across passage sets. Figure 1 provides graphic presentations of the results.

The primary dependent measures of interest were linear estimates of weekly growth along with the corresponding weekly *SEb* and *SEE*. Marginal mean estimates for weekly growth approximated 1.36, 1.79, and 0.45 WRCM gain per week for FAIP-R, AIMSweb, and DIBELS passage sets, respectively. Differences between weekly growth estimates across passage sets were within the range

Table 1
Descriptive Statistics for Progress Monitoring Outcomes Across Three Passage Sets: Intercept, Weekly Growth, and Standard Error

Statistic	Passage Set	*M*	*SD*	Skewness	Kurtosis
Intercept					
	FAIP-R	84.81	29.92	0.01	1.00
	AIMSweb	94.29	33.14	0.06	1.21
	DIBELS	100.40	30.62	0.12	1.17
Slope					
	FAIP-R	1.36	1.15	0.48	0.71
	AIMSweb	1.79	1.35	0.52	0.62
	DIBELS	0.45	1.12	−0.13	0.09
SEb					
	FAIP-R	0.64	0.20	1.72	5.29
	AIMSweb	0.71	0.23	1.17	2.44
	DIBELS	0.91	0.29	0.91	1.03
SEE					
	FAIP-R	10.68	2.96	1.34	3.04
	AIMSweb	11.89	3.29	1.14	2.78
	DIBELS	15.26	4.27	0.86	0.84

Note. FAIP-R = Formative Assessment Instrumentation and Procedures for Reading; DIBELS = Dynamic Indicators of Basic Early Literacy Skills; *SEb* = standard error of slopes; *SEE* = standard error of the estimate. *n* (progress monitoring data set) = 68 within each probe set with a mean (*SD*) of 19.91 (0.41) number of observations per data set across a mean (*SD*) of 11.66 (0.77) weeks.

of 0.43 and 1.34. Paired contrasts for estimates of weekly growth yielded statistically significant differences across passage sets ($p < .05$; Table 2).

The *SEb* and *SEE* are the most useful dependent measures to evaluate passage sets. Marginal mean estimates for the weekly *SEb* were 0.64, 0.71, and 0.91 WRCM gain per week for FAIP-R, AIMSweb, and DIBELS, respectively. Differences between marginal means were within the range of −0.28 and 0.08; and all paired contrasts yielded statistically significant differences across passage sets (Table 2). The rank order of passage sets from the smallest to largest magnitudes for weekly *SEb* was FAIP-R, AIMSweb, and DIBELS (Figure 1). Similar results were observed for *SEE* with marginal means within the range of approximately 11, 12, and 15 WRCM for FAIP-R, AIMSweb, and DIBELS, respectively (Table 2). Differences across passage sets were within the range of −4.58 and 1.21, and all paired contrasts of marginal means yielded statistically significant differences. The rank order of passage sets from the smallest to largest magnitudes for *SEE* was identical to that of weekly *SEb* (i.e., FAIP-R, AIMSweb, DIBELS). Figures 2 and 3 present case examples selected from our data that illustrate differences in estimated slope across passage sets for individual students (Figure 2) as well as magnitude of error observed when monitoring individual student progress (Figure 3).

Despite the inconsistencies between growth estimates across passage sets, the correlations of student performances across passage sets were consistently large in magnitude and statistically significant ($p < .01$; Table 3). The intercepts associated with the linear estimates of weekly student growth correlated at or above .98 across passage sets; moreover, the estimates of weekly growth approximated or exceeded 0.60. It is interesting to note that analysis of the standard error yielded strong and statistically significant correlations (Table

Table 2
Planned Pairwise Comparisons of Intercept, Weekly Growth, and Standard Error: Differences in Progress Monitoring Outcomes Across Three Passage Sets

		Marginal Mean		Pairwise Comparisons[a,b]	
Statistic	Passage Set	Mean	Standard Error	AIMSweb $M_r - M_c$	DIBELS $M_r - M_c$
Intercept					
	FAIP-R	84.81	3.63	−9.48**	−15.59**
	AIMSweb	94.29	4.02	—	−6.11**
	DIBELS	100.40	3.71	—	—
Slope					
	FAIP-R	1.36	0.14	−.43**	.91**
	AIMSweb	1.79	0.16	—	1.34**
	DIBELS	0.45	0.14	—	—
SEb					
	FAIP-R	0.64	0.02	−0.08**	−.28**
	AIMSweb	0.71	0.03	—	−.20**
	DIBELS	0.91	0.04	—	—
SEE					
	FAIP-R	10.68	0.36	−1.21**	−4.58**
	AIMSweb	11.89	0.40	—	−3.36**
	DIBELS	15.26	0.52	—	—

Note. FAIP-R = Formative Assessment Instrumentation and Procedures for Reading; DIBELS = Dynamic Indicators of Basic Early Literacy Skills; *SEb* = standard error of slopes; *SEE* = standard error of the estimate.
[a]Difference between marginal means subtracting the column value from the row values ($M_r - M_c$).
[b]Least significant difference adjustment for multiple comparisons of marginal mean differences.
*$p < .05$.
**$p < .01$.

3). That result indicates that the relative standing of individual students is reliable and consistent in terms of the standard error associated with their progress monitoring outcomes. Correlations for *SEb* were in the range of .62 and .74; and *SEE* correlations were in the range of .55 to .67.

The observed correlations associated with standard error were unexpected, so we conducted exploratory analysis to examine the phenomenon. A review of the larger correlation matrix (*SE*, *SEb*, intercept, slope) indicated that the initial level of student performance and/or the intercept of progress monitoring trend estimates were both related to the standard error. Correlation coefficients were statistically significant ($p < .01$) with magnitudes within the moderate range (.45 to .60). On further review, it was apparent that the magnitude of standard errors was generally less among students with lower oral reading rates and greater among students with higher reading rates. The relationship between intercept and standard errors was therefore examined and explicated with post hoc analysis.

Partial correlations were used for post hoc analyses. The intercepts for FAIP-R, AIMSweb, and DIBELS were entered as control variables. We reasoned that the intercept was a convenient control to reduce the influences of prior achievement and inconsistent reading rates among students. In turn, we expected that analyses using partial correlations might help isolate the relationship and relative

consistency of progress monitoring outcomes across passage sets. As depicted in Table 3, *partial* correlations (range = .81 to .87) were more robust for estimates of growth across passage sets. That is, the outcomes of progress monitoring appear more consistent and reliable across alternate passage sets when the intercept was used to control for initial level of student performances. In contrast, both the weekly *SEb* (range = .47 to .64) and *SEE* (range = .31 to .48) were both somewhat less consistent for students across passage sets when the intercept was used to control for initial level of student performance.

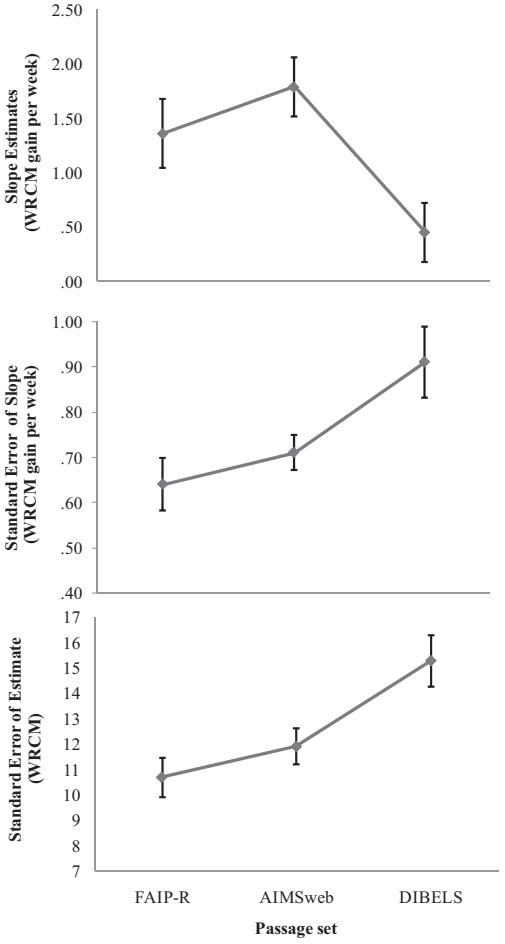


Figure 1. Marginal means and standard errors (95% confidence interval) for each dependent measure and passage set are depicted. From top to bottom, the panels present intercept, weekly growth estimates of growth, standard error for weekly growth estimates, and standard error of the estimate. WRCM = words read correctly per minute; FAIP-R = Formative Assessment Instrumentation and Procedures for Reading; AIMSweb; DIBELS = Dynamic Indicators of Basic Early Literacy Skills.

The relationships of intercepts, slopes, and standard errors led us to examine growth estimates within specified ranges of initial achievement, which were defined by the intercept (i.e., estimate of the initial reading rate when progress monitoring began). Four ranges of 40 WRCM were identified that generally correspond to expected achievement in first, second, third, and fourth grades (Shapiro, 1996). The corresponding estimates of standard error can be used to guide interpretation of progress monitoring data for students whose intercept (or baseline level) is likely to approximate a particular range. Table 4 presents observed levels of standard error disaggregated into four general levels of oral reading (i.e., 0–40, 41–80, 81–120, 121+). Across the three passage sets, there is an increasing trend in mean *SEb* and *SEE* that corresponds to the increasing rates of students' initial fluency. Results within each of the four specified ranges of initial fluency correspond with the overall observed levels of measurement error, with the FAIP-R passage set resulting in the least amount of error and the DIBELS passage set resulting in the greatest amount of error.

Discussion

Extant research has almost exclusively evaluated the reliability, validity, and sensitivity of the CBM-R data at the group level (e.g., Ardoin & Christ, 2008; Ardoin et al., 2004; Hintze & Christ, 2004). Although studies have repeatedly demonstrated the strength of CBM-R procedures for evaluating students' relative level and rates of growth, CBM-R data are also being used to evaluate the progress of individual students and therefore

must be evaluated for this purpose. When the data are interpreted at the individual student level, a student's performance is not compared relative to other students but rather to the individual student's previous performance(s). Recent studies have suggested that there may be problems in analyzing CBM-R progress monitoring data at the individual student level

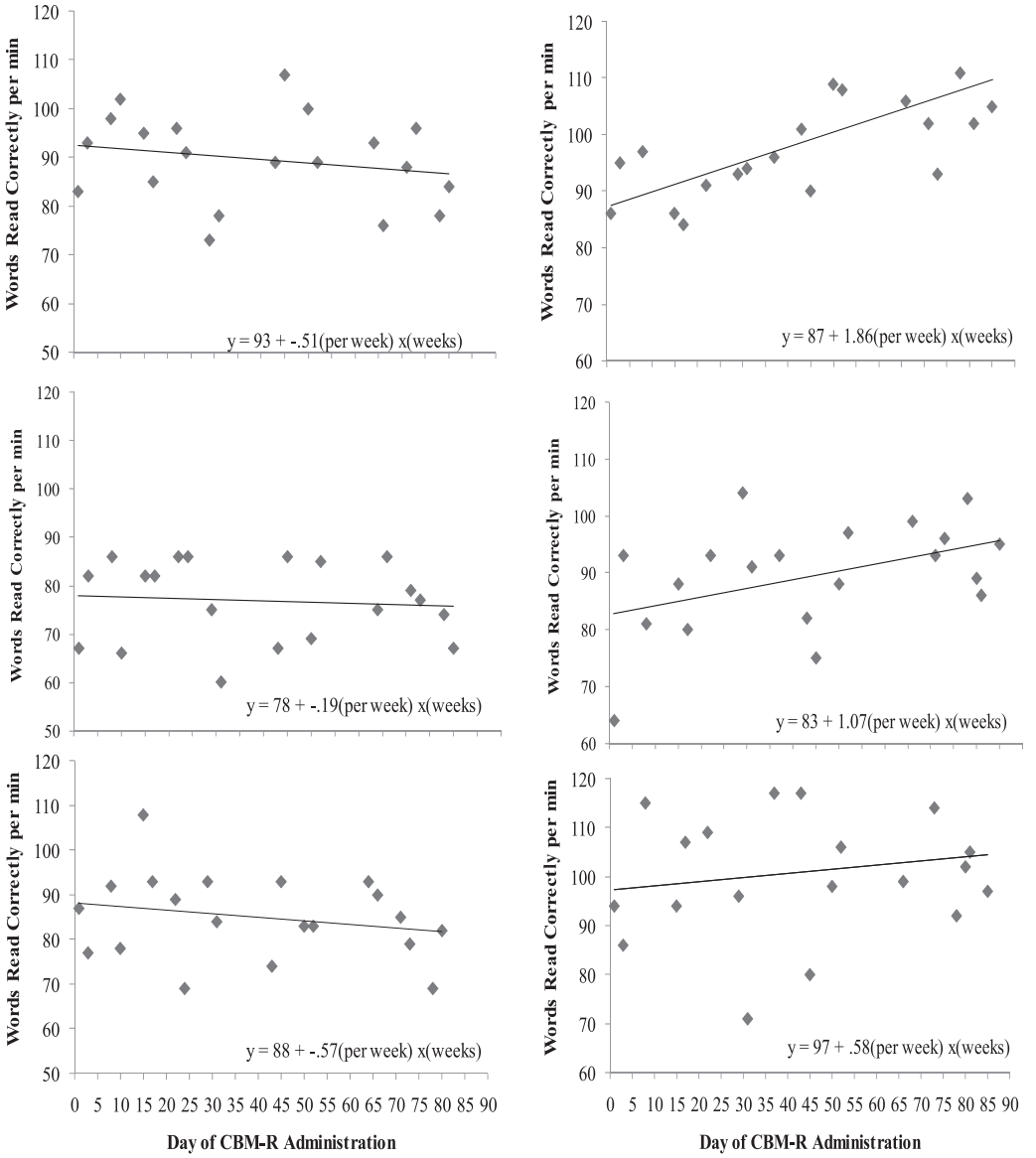


Figure 2. Two case examples of progress monitoring data and estimates of weekly growth across three passage sets. From top to bottom, the panels present AIMSweb, Formative Assessment Instrumentation and Procedures for Reading, and Dynamic Indicators of Basic Early Literacy Skills passage set examples for one case that illustrates negative weekly growth (left panels) and another that illustrates positive weekly growth (right panels). CBM-R = curriculum-based measurement for oral reading.

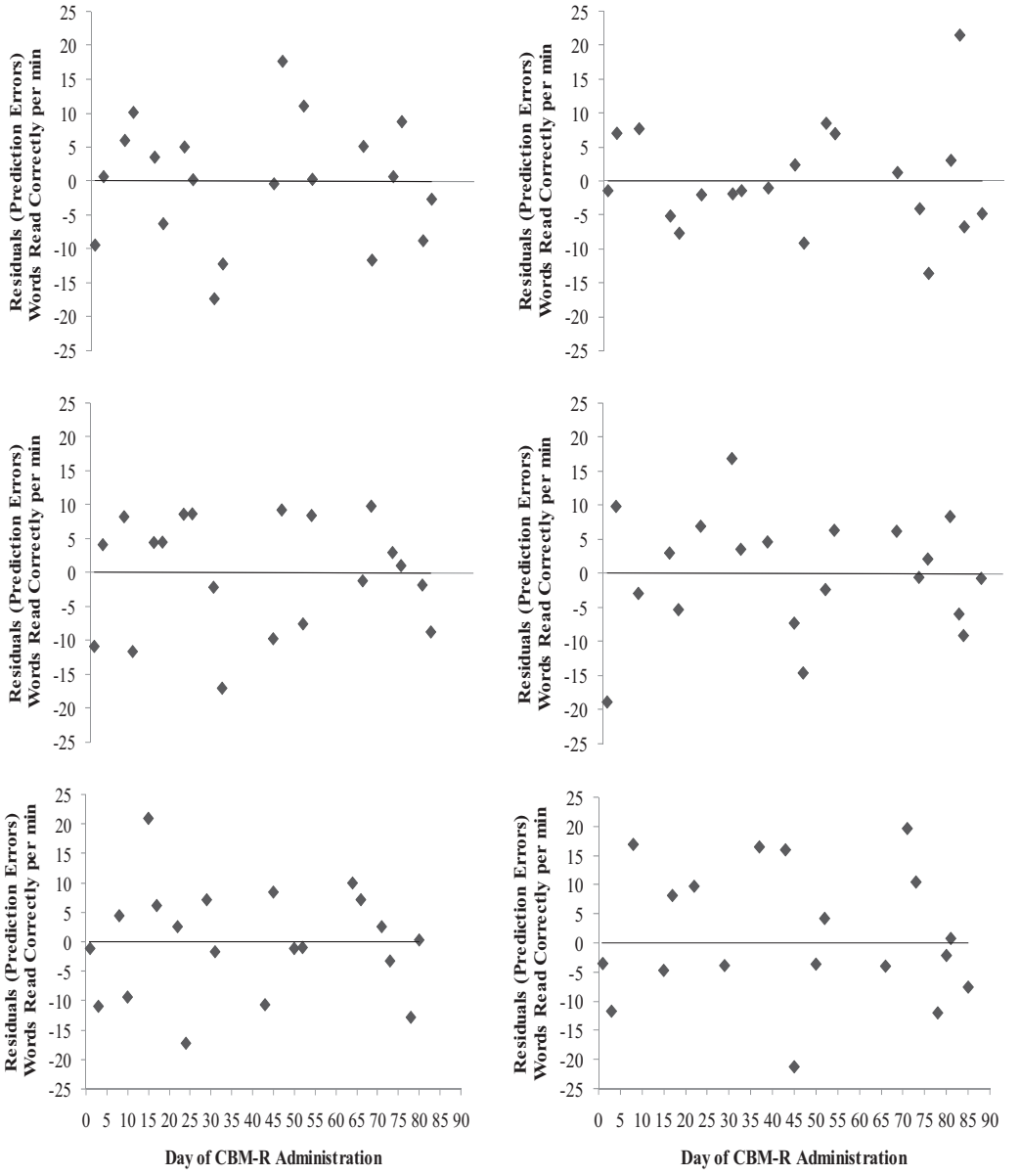


Figure 3. Unstandardized residuals (i.e., error of the estimates) for two cases examples of progress monitoring data. As in Figure 2, the panels present from top to bottom data for AIMSweb, Formative Assessment Instrumentation and Procedures for Reading, and Dynamic Indicators of Basic Early Literacy Skills Dynamic Indicators of Basic Early Literacy Skills passage set examples for one case with negative weekly growth (left panels) and another that illustrates positive weekly growth (right panels). CBM-R = curriculum-based measurement for oral reading.

Table 3
Consistency of Progress Monitoring Outcomes Across Three Passage Sets: Planned Correlations and Post Hoc Partial Correlations With Intercept as a Controlling Variable

	Correlations Across Passage Sets		Partial Correlations Using Intercept, Passage Sets	
Passage Set	AIMSweb $r_{x,y}$	DIBELS $r_{x,y}$	AIMSweb $r_{x,y:\text{intercepts}}$	DIBELS $r_{x,y:\text{intercepts}}$
Intercept				
FAIP-R	.98	.98		
AIMSweb	—	.98		
DIBELS	—	—		
Weekly growth (Slope)				
FAIP-R	.70	.59	.81	.81
AIMSweb	—	.68	—	.87
DIBELS	—	—	—	—
SEb				
FAIP-R	.64	.62	.52	.47
AIMSweb		.74	—	.64
DIBELS	—	—	—	—
SEE				
FAIP-R	.55	.56	.35	.31
AIMSweb		.67	—	.48
DIBELS	—	—	—	—

Note. DIBELS = Dynamic Indicators of Basic Early Literacy Skills; FAIP-R = Formative Assessment Instrumentation and Procedures for Reading; *SEb* = standard error of slopes; *SEE* = standard error of the estimate. Partial correlations were run post hoc with three intercepts as controlling variables (AIMSweb, FAIP-R, and DIBELS) to mediate the inconsistencies of oral reading skills among students; $n = 68$ within each passage set; all coefficients were statistically significant ($p < .01$).

because of excessive measurement error, which has been attributed in large part to the variability in the difficulty of passages within passage sets. Using extant studies, Christ (2006) estimated that error resulted in SEb larger than expected rates of student growth, making the interpretation of individual students' rates of growth using CBM-R progress monitoring procedures difficult.

The primary purpose of the current article, therefore, was to evaluate the extent to which the FAIP-R passage set, comprised of passages selected as equivalent based on field testing of a larger set of passages, would result in reduced measurement error as compared to two published passage sets. Results confirmed our hypothesis that the Euclidean distance method described by Christ and Ardoin (2009) results in an improved set of parallel progress monitoring passages. Both indices suggested that the FAIP-R passage set resulted in significantly less measurement error than did the AIMSweb and DIBELS passage sets. Additional evidence of the benefits of estimating passage difficulty using students' reading of passages as opposed to readability formulas is provided by the fact that the AIMSweb passage set resulted in significantly less measurement error as compared to the DIBELS passage set. Although the difficulty of the AIMSweb passage was determined primarily by readability formulas, a small sample of students' readings were used to eliminate some passages (Howe & Shinn, 2002). In contrast,

Table 4
Standard Errors of Weekly Growth and Standard Errors of the Estimate After Progress Monitoring with Curriculum-Based Measurements of Oral Reading: Disaggregated by Range of Initial Levels of Student Performances

	Subsets by Intercept Range									
	0–40 $n = 5$		41–80 $n = 12$		81–120 $n = 40$		121+ $n = 11$		Total $n = 68$	
Passage Set	*M*	(*SE*)	*M*	(*SE*)	*M*	(*SE*)	*M*	(*SE*)	*M*	(*SE*)
Mean intercept	28	(7)	66	(10)	97	(10)	139	(22)	93	(31)
SEb										
FAIP-R	0.49	(0.04)	0.55	(0.03)	0.65	(0.03)	0.75	(0.06)	0.64	(0.02)
AIMSweb	0.56	(0.03)	0.65	(0.05)	0.71	(0.03)	0.87	(0.10)	0.71	(0.03)
DIBELS	0.69	(0.07)	0.75	(0.06)	0.91	(0.04)	1.23	(0.09)	0.91	(0.04)
SEE										
FAIP-R	7.7	(0.72)	9.1	(0.41)	11.0	(0.46)	12.5	(0.99)	10.7	(0.36)
AIMSweb	8.8	(0.64)	10.6	(0.63)	12.0	(0.44)	14.4	(1.45)	11.9	(0.4)
DIBELS	11.0	(1.26)	12.4	(0.77)	15.3	(0.54)	20.3	(1.32)	15.3	(0.52)

Note. SEb = standard error of slopes; FAIP-R = Formative Assessment Instrumentation and Procedures for Reading; DIBELS = Dynamic Indicators of Basic Early Literacy Skills; *SEE* = standard error of the estimate. The cut points for intercept ranges were selected because they are likely to correspond with student performances in first, second, third, and fourth grades, respectively.

DIBELS passages were selected strictly by readability formulas (Good & Kaminski, 2002).

A second purpose of the current study was to evaluate differences in AIMSweb and DIBELS passage sets. Despite their wide use for making educational decisions for elementary students, there is no existing study that has contrasted these two CBM-R progress monitoring passage sets. Results of this study suggest that there in fact might be substantial differences between the quality of these passage sets. The AIMSweb passage set resulted in significantly smaller *SEb* and *SEE*. Although measurement error is not likely to be the first thing that schools consider when determining which of these two commercially available passage sets to adopt, we believe it actually should be of utmost concern for two reasons. First, the passage set with the larger *SEE* means that when a student's data are plotted on a graph, there is going to be substantial variability in the student's performance even in the absence of learning. In the collection of our data, there were students whose third or fourth to last data point was below their baseline data points, replicating findings of a recent study by Francis et al. (2008). When data collected weeks after intervention implementation fall below baseline data, it may be more difficult (and rightly should be) to convince teachers and/or parents that a child is making progress (e.g., Figure 2). The second reason that schools should consider measurement error is the accuracy of the decisions that can be drawn from progress monitoring data. Passage sets that produce larger *SEb* will result in a greater number of false positive and false negative decisions regarding students' response to intervention, meaning that districts will place more students in special education who do not need it and refuse special education services to students who actually need it.

Another interesting difference found between the three passage sets was the signifi-

cant differences in intercepts, with the FAIP-R passage set having the lowest intercept and the DIBELS' intercept being the greatest. This finding has several important implications for schools. First, these results suggest that schools should not compare a student's performance across these two sets of passages. Thus, if a student transfers from one school using DIBELS passages to monitor progress to a school using AIMSweb passages to monitor progress, the student's growth should not be estimated using his or her performance on passages administered at the other school. Second, given that the post hoc analyses indicated that measurement error is greater when monitoring the performance of students with higher levels of fluency, schools may wish to consider the appropriate level at which to monitor students' performance. Although one might attribute the higher levels of measurement error of the DIBELS passage set to students having read those passages at a greater rate of fluency, descriptive statistics presented in Table 4 indicate that across the four subsets of initial fluency levels, error rates were consistently higher for the DIBELS passage set.

Post hoc analyses indicating that measurement error increased as students' initial fluency level increased suggest the potential need for researchers to once again address the question of the most appropriate level at which to monitor student progress. Although it is not clear that a definitive answer was ever drawn regarding whether students should be monitored at their instructional or frustration level, studies examining factors that may contribute to the sensitivity of CBM-R have failed to adequately examine data at the individual child level (Dunn & Eckert, 2002; Fuchs & Deno, 1992; Hintze, Shapiro, & Daly III, 1998; Shinn, Gleason, & Tindal, 1989). Results of the current study suggest that future studies should examine the reliability and accuracy of decisions regarding students' response to instruction as a function of the difficulty of passages used to monitor progress.

Despite the proportionately large indices of measurement error observed across passage sets, significant differences in measurement error between passage sets, and significant differences in intercept between passage sets, there was relative consistency in students' level of performance and rate of growth across passage sets (Table 3). Evidence that students' relative level of performance and rates of growth are consistent across CBM-R passage sets is consistent with previous studies (Fuchs & Deno, 1992; Hintze et al., 1998; Shinn et al., 1989), which examined the traditional psychometrics of CBM-R. However, these studies did not examine the quality of CBM-R for evaluating individual student growth. Results of this study confirm a recommendation that has been made by leaders within CBM-R research that has been largely ignored, which is the need to examine CBM-R outside of the scope of traditional reliability and validity (Deno et al., 2001; Francis et al., 1994; Fuchs & Fuchs, 2002). Results of CBM-R progress monitoring data are used for purposes other than examining a student's relative performance. CBM-R progress monitoring data are being used for making absolute comparisons. Students' performances at baseline are being compared to their performance after as few as 4 weeks of intervention, and these comparisons are being used to make critical educational decisions for individual students. The reliability and accuracy of decisions regarding individual student gains across the three passage sets was not evaluated as part of this study. Future studies need to address this question along with alternative procedures for increasing the reliability and accuracy of decisions made using CBM-R progress monitoring data.

Limitations

Multiple limitations are associated with results of the current study. First, the generalizability of the findings are limited given that only second- and third-grade students participated and all participants were located in the southeastern region of the United States. A second limitation is that a small portion of the third-grade students ($n \sim 8$) enrolled in School A also participated in the field testing study of the FAIP- R passages (Christ & Ardoin, 2009). Thus, the selection of the parallel pas-

sages was at least partially a function of their previous reading of these passages. A third, related limitation is that all of the students enrolled in School A attended the same school as those students who participated in the Christ and Ardoin study. The reduction in observed measurement in the FAIP-R passages as compared to the other two passage sets may be a function of similarities between students who field tested the passages and the participants of the current study. Thus, greater rates of error may be observed when using FAIP-R passages to monitor students exposed to different curricula. Greater rates of error might also be observed across passage sets in nonexperimental conditions (i.e., typical progress monitoring conditions). Although we cannot claim that for each student the same experimenter always collected his or her data, experimenters were assigned to particular students and assessed their assigned students except on make-up occasion or experimenter absences. However, regardless of experimenter, administration directions and procedures were consistent across sessions, given the use of written protocols and assessment of procedural integrity. We also employed what was believed to be an incentive ($0.05 for beating predetermined score) for students to read their best during each session. It is possible that this procedure influenced the results; however, it was implemented consistently across passage sets. Nevertheless, giving an incentive for exceeding a predetermined score is a deviation from standard protocol, although we have observed that many teachers employ similar tactics in practice.

Another limitation of the current study is, regardless of grade level and reading ability, all students were monitored with the same three sets of third-grade probes. This decision was made primarily because we only had access to third-grade FAIP-R passages. We had not expected, however, the vast difference in intercepts across passage sets. Failing to monitor the progress of the more skilled students with passages of greater difficulty may have artificially increased observed levels of measurement error. Examination of Table 4, however, indicates that across reading levels mean rates of error were at proportionately high levels regardless of students' initial level of performances. Mean *SEE* were also consistent with previously reported *SEE* (Christ, 2006).

Conclusion

Despite evidence that the procedures employed to develop a set of parallel passages significantly reduced observed measurement error, the findings remain a bit concerning. The average *SEb* for the FAIP-R passage set ($M = 0.64$) was large in proportion to the expected 1–2 WRCM gain per week (Deno et al., 2001). Using the observed weekly *SEb* for FAIP-R passages to calculate a 68% confidence interval ($1.68 \times 0.64 = 1.08$) around an observed rate of growth of 1.5 would suggest that a student's response to instruction was somewhere between inadequate ($1.36 - 1.08 = .28$) to excellent ($1.36 + 1.08 = 2.44$). The obtained mean weekly *SEb* for the AIMSweb (68% confidence interval: 1.5 ± 1.19) and DIBELS (68% confidence interval: 1.5 ± 1.53,) passages sets would result in even less clarity, with the confidence interval for the DIBELS data actually suggesting the possibility that the student's performance decreased significantly. The scientific implications of these findings is that we must further evaluate CBM-R for all purposes for which it is being used, which means we critically examine whether reliable decisions can be made regarding individual student's response to instruction. Furthermore, developers of passage sets must provide indices of measurement error regarding student level and rate of growth as well as the reliability and accuracy of estimated rates of growth at the individual child level. Researchers should also explore alternative techniques for reducing error such as the administration of multiple passages per session and the equating of passages using statistical procedures. Unfortunately, evaluation of procedures for developing equivalent level passages is in its infancy, and replications and extensions of this and other procedures (e.g., equipercentile equating; Francis et al., 2008) for reducing measurement error in CBM-R progress monitoring are needed.

These data also have important implications for schools. First and most important, these findings, along with the results of other studies (Betts, Pickard, & Heistad, 2009; Christ, 2006; Christ & Ardoin, 2009; Francis et al., 2008; Poncy et al., 2005), indicate that educational decisions must not be made using CBM-R progress monitoring data as the primary outcome measure. As it currently stands, there are no well-validated measures for monitoring student progress across short periods of time (e.g., 0–15 weeks); thus schools should consider multiple sources of data including but not limited to universal screening and subskill mastery data. Second, schools must acknowledge that in all measurement devices error is present. Each observation collected on a student represents the student's true skills plus error. CBM-R data being used to assess a student's level and rate of growth are no exception and must be treated as only estimates. Schools should therefore report confidence intervals around obtained estimates of level and rates of growth (Christ & Coolong-Chaffin, 2007). In addition, schools paying for access to CBM-R probes or web-based services for monitoring their students' progress should insist that indices of measurement error at the individual child level be reported.

Finally, CBM-R progress monitoring procedures and passage sets were not initially developed with the intention of being used for making special education eligibility decisions. Now that CBM-R procedures are being used for such purposes, we must examine the quality of every passage set as if it is a new standardized test. Evidence that one passage set is reliable, valid, or sensitive in no way provides evidence of the quality of other passage sets. Just as a study demonstrating the construct validity of one published norm-referenced reading achievement test does not provide construct validity for all such tests, supporting evidence for one set of CBM or other general outcome measure does not provide evidence for all general outcome measures in that family. We must encourage publishers to provide technical manuals specific to each passage set regarding its reliability, validity, accuracy, and sensitivity for all purposes for which their instrumentation are intended to be used. Such technical manuals would allow practitioners to know the confidence with which they can interpret the data they have collected regarding a student's response to intervention.

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